KareShafaa

Handover package

Future Health Challenge 05 – closing the loop on abnormal screening results

Problem brief · runnable project · task agreement · test results · pilot plan · handover session

Prepared 04 November 2026. All data synthetic. Values marked ASSUMPTION or CONFIRM are not yet agreed.

KareShafaa handover package – Challenge 05

Everything needed to understand, run, evaluate and pilot KareShafaa. All data is synthetic. Printable version: `KareShafaa_Handover_Package.pdf` (also `.docx`).

#	Part	File
1	Problem brief	problem_brief.md
2	Runnable project (setup, env vars, one-command example)	runnable_project.md, `scripts/handover_example.sh`
3	How to run the task (input / output agreement)	how_to_run_the_task.md, examples/
4	Test results	test_results.md
5	Pilot plan	pilot_plan.md (condensed from `docs/pilot_plan.md`)
6	Handover session	handover_session.md

Six-part checklist

- [x] **1. Problem brief** – users, task, scope in/out, intended outcome → `problem_brief.md`
- [x] **2. Runnable project** – clean-clone setup (SQLite and Docker/Postgres), all env vars with secrets named only, repeatable one-command example → `runnable_project.md`, `scripts/handover_example.sh`, `scripts/handover_example.py`
- [x] **3. How to run the task** – inputs (HL7 ORU, FHIR) with fields; outputs (alerts, tiers, SLA timers, FHIR, audit); sample inputs and the example run record → `how_to_run_the_task.md`, `examples/input_oru_r01_fit_positive.hl7`, `examples/input_fhir_bundle_hba1c.json`, `examples/example_run_record.md`, `examples/example_run_record.json`, `examples/example_output_fhir_bundle_case.json`
- [x] **4. Test results** – test cases and counts, baseline comparisons, errors found, known limits → `test_results.md`
- [x] **5. Pilot plan** – participants, success criteria, review lead role, review dates → `pilot_plan.md`
- [x] **6. Handover session** – 60-minute agenda, results review, walkthrough, Q&A, decisions, next steps, sign-off table → `handover_session.md`

1. Problem brief – Challenge 05: closing the loop on abnormal screening results

The problem

When a screening test (bowel FIT, cervical HPV/cytology, breast mammography, diabetes HbA1c/fasting glucose) comes back abnormal, the patient must be told, understand it, and reach the next step (confirmatory test, specialist, treatment). Each hand-off can fail silently. In Abu Dhabi's colorectal screening programme (2015–16), only **30.5 % of people with a positive FIT had a colonoscopy** (BMC Cancer 2021, https://link.springer.com/article/10.1186/s12885-021-08623-9). Longer delay after a positive FIT is linked to higher odds of cancer and of advanced-stage disease (Corley et al., JAMA 2017, https://jamanetwork.com/journals/jama/fullarticle/2620087). Sources and the pilot sizing are in `docs/pilot_plan.md`.

Users

User	What they need from KareShafaa
Screening coordinator	One work queue of abnormal results, ordered by urgency, with an owner, a deadline (SLA timer) and the next action for each
Family doctor (GP)	Their patients' abnormal results, what was decided, what is still open
Specialist / receiving service	Referrals to accept or reject with a reason; appointment outcomes
Programme clinical lead	Alerts when steps or pathway targets are missed; loop-closure, timeliness and equity reports; rule and model governance
Administrator	Accounts, roles, integrations status, audit chain
Patient	Their result explained in Arabic, English, Hindi or Urdu; confirm or ask to change an appointment; say what gets in the way

The task the system performs

Take an abnormal screening result (HL7 v2 ORU^R01 or FHIR R4 Bundle), match it to the right patient, apply the clinical rules to set an urgency tier and pathway with a target date, open a tracked case with an owner and a step-by-step SLA clock, escalate when a step or the pathway target is missed, record every action in a hash-chained audit log, and close the loop only with evidence (attended visit or a signed clinical decision).

Scope

In: the four screening programmes above; inbound HL7 v2 and FHIR; rules engine with human review; case workflow and SLA escalation ladder (L1 reminder to L4 director); patient app in four languages; FHIR R4 export; outcome and equity reporting; ML outreach-priority model that only orders cases *within* a tier; RBAC and audit.

Out (not built, or simulated): real SMS/WhatsApp gateway (messages are simulated); live Malaffi or laboratory connection; single sign-on and MFA; billing; diagnosis or treatment decisions (always made by clinicians); real patient data (all data in this repository is synthetic).

Intended outcome

More abnormal results reach their next clinical step, sooner and within target, with no group left behind and with evidence for every closure. Pilot measures and targets are in `pilot_plan.md` (targets are labelled ASSUMPTION until agreed with the partner).

2. Runnable project

Repository root: this `app/` directory. Python 3.12+ (CI uses 3.13). All data is synthetic.

A. Local, SQLite (quickest)

```
git clone <repo-url> kareshafaa && cd kareshafaa
python3 -m venv .venv && .venv/bin/pip install -r requirements-dev.txt
.venv/bin/python -m kareshafaa.seed --patients 2000 --seed 20261120     # fresh synthetic DB i
n data/
.venv/bin/python -m kareshafaa.serve --host 127.0.0.1 --port 8000 --workers 1
# open http://127.0.0.1:8000  - outside public mode, first passwords for the staff accounts ar
e generated at random
# and written to the file named by KARESHAFAA_DEMO_CREDENTIALS_FILE (mode 0600, never committe
d)
```

Tests: `.venv/bin/python -m pytest -q` (builds its own throwaway database in a temp dir).

B. Docker + PostgreSQL

```
cp .env.example .env   # or create .env with the three secrets below; never commit it
docker compose up --build    # app on http://localhost:8000, Postgres 16 in volume "pgdata"
```

`docker-compose.yml` reads the secrets `KARESHAFAA_SECRET_KEY`, `KARESHAFAA_DEMO_PASSWORD` and `POSTGRES_PASSWORD` from `.env`. The container runs `python -m kareshafaa.bootstrap` (creates missing tables and columns, additive only) before starting the server.

C. One-command example (fresh database, fixed seed, run record)

```
bash scripts/handover_example.sh [OUT_DIR]     # default ./handover_run ; about 15 s after ins
tall
```

It creates `.venv` if missing, builds a new SQLite database with 300 synthetic patients (seed 2026), sends `examples/input_oru_r01_fit_positive.hl7` and `examples/input_fhir_bundle_hba1c.json` through the real API, records alert, tier, pathway target, SLA timer, the escalation sweep at +1 and +40 days, validates the FHIR export and the audit chain, runs the rules engine against the independent reference on 20,000 randomised results (seed 20262020) and writes `OUT_DIR/run_record.json` and `OUT_DIR/run_record.md`. The output committed in `examples/example_run_record.*` came from this script. Verified from a fresh `git clone` in a temp dir (see `test_results.md`). Times and case ids depend on the run date; decisions, tiers and checks do not.

Environment variables

All settings use the prefix `KARESHAFAA_` (see `kareshafaa/config.py`); defaults suit local development. **Secrets are referenced by name only – set them in `.env` or the hosting dashboard, never in git.**

Variable	Secret?	Purpose
`KARESHAFAA_SECRET_KEY`	**secret**	Signing key for sessions and tokens (64 random characters)
`KARESHAFAA_DEMO_PASSWORD`	**secret**	First password for the provisioned staff accounts (internal variable name)
`DATABASE_URL` / `KARESHAFAA_DATABASE_URL`	**secret** when it holds a password	Database; default SQLite file in `data/`
`POSTGRES_PASSWORD`	**secret**	Docker Compose database password
`KARESHAFAA_MODE`	no	`standard`, `production` or `public_demo` (internal identifier)
`KARESHAFAA_ENV`	no	`dev`, `test`, `prod`
`KARESHAFAA_DEMO_LOGINS`	no	Role-switcher logins; **must be** `false` outside test environments
`KARESHAFAA_PUBLIC_DEMO`	no	Public sandbox with nightly reset (internal identifier)
`KARESHAFAA_DEMO_CREDENTIALS_FILE`	path only	Where generated first passwords are written (0600)
`KARESHAFAA_DATA_DIR`, `KARESHAFAA_MODEL_DIR`	no	Data and model folders
`KARESHAFAA_SWEEP_INTERVAL_SECONDS`	no	Background SLA/escalation sweep (300; 0 = off)
`KARESHAFAA_DATA_STEPS`	no	Start-up data steps: `background` or `off`
`KARESHAFAA_SESSION_IDLE_MINUTES`, `KARESHAFAA_TOKEN_TTL_MINUTES`	no	Session expiry
`KARESHAFAA_LOGIN_MAX_FAILURES`, `..._LOGIN_WINDOW_MINUTES`, `..._LOGIN_LOCKOUT_MINUTES`	no	Lockout
`KARESHAFAA_COOKIE_SECURE`	no	Force the Secure cookie flag (auto by default)
`KARESHAFAA_SUPPRESSION_THRESHOLD`	no	Small-cell suppression in reports (5)
`KARESHAFAA_MAX_INFLIGHT_REQUESTS`, `KARESHAFAA_DB_POOL_SIZE`	no	Load shedding and pool size
`KARESHAFAA_SLA_CONFIG_PATH`	no	Optional JSON overriding default SLA hours
`WEB_WORKERS`	no	Server worker processes
`KARESHAFAA_BOOTSTRAP_SOURCE`	no	SQLite file copied into an empty Postgres on first start

Other tuning values (ML gates, fairness tolerances, escalation fractions) are listed with comments in `kareshafaa/config.py`.

3. How to run the task: input / output agreement

The task: **an abnormal screening result goes in; a tracked, owned, timed case comes out, and stays visible until the loop is closed with evidence.** All examples below use synthetic data.

Inputs

Channel	Endpoint	Permission	Example file
HL7 v2.5.1 ORU^R01 (pipe-delimited, `\r` segment separator, UTF-8, ≤ size limit)	`POST /api/ingest/hl7v2`	`case:create`	`examples/input_oru_r01_fit_positive.hl7`
FHIR R4 Bundle (JSON; `message` or `collection`)	`POST /api/ingest/fhir`	`case:create`	`examples/input_fhir_bundle_hba1c.json`
FHIR R4 history Bundle (Encounter, Medication*, AllergyIntolerance, Condition) for an existing patient	`POST /api/ingest/fhir-history`	`record:write`	–
Manual entry by a coordinator	`POST /api/results`	`case:create`	–

HL7 v2 fields read (`kareshafaa/services/ingest.py: parse_hl7v2`)

Segment-field	Use
MSH-1/MSH-2	Delimiters (read from the message, never assumed)
MSH-9	Must be `ORU^R01`
MSH-10	Message control id – idempotency key, echoed in the ACK
PID-3 (all repetitions)	Patient identifiers (first = primary; others = alternates). Matching is on identifiers only
PID-5, PID-7, PID-8, PID-13	Name, date of birth, sex, phone – a birth-year mismatch sends the message to human match review; demographics only pre-fill a new registration, never overwrite
OBR / ORC-12 / OBR-16, ORC-21 / PV1-3	Specimen time, ordering clinician, screening site
OBX-2, OBX-3, OBX-5, OBX-6, OBX-11	Value type, LOINC code, value (incl. `SN` comparators), unit, result status (P/F/C)

FHIR fields read (`parse_fhir_bundle`)

`Bundle.identifier.value` (idempotency key) · `Patient.identifier[].value`, `Patient.birthDate` · `Observation.code.coding[].code` (LOINC), `Observation.valueQuantity.value/unit` or `valueString` / `valueCodeableConcept.text`, `Observation.status` · `ServiceRequest.requester` / `locationReference`.

Recognised LOINC codes

Code	Programme / test
29771-3	Bowel (colorectal) – FIT

Code	Programme / test
77379-6 / 10524-7	Cervical – HPV / cytology
24606-6	Breast – mammography (BI-RADS)
4548-4, 59261-8	Diabetes – HbA1c (%, mmol/mol)
1558-6, 14771-0	Diabetes – fasting plasma glucose

Unknown codes, malformed values, unit mismatches, preliminary results and identity doubts are **never dropped**: they are kept and routed to a human review queue (fail-safe; see `docs/clinical_safety_case.md`).

Outputs

Output	Where	Notes
ACK (HL7)	response `ack`	`MSA|AA` accepted, `MSA|AE` kept for review (with the reason)
Ingest status	response `status`	`processed`, `resolved` (after human review), `needs_match_review`, `needs_review`
Alert decision / case	response `case`, `GET /api/cases/{id}`	rule id, programme, **tier** (T0 critical, T1 urgent, T2 high, T3 routine), pathway, pathway target date, human-review flag and reasons, "why this priority" explanation
SLA timers	case `steps[]`, `sla`	Each state has its own clock per tier (`kareshafaa/sla.py`, e.g. *issued*: 1 / 4 / 24 / 48 h for T0–T3; illustrative defaults, CONFIRM). T2/T3 count UAE working days
Escalation alerts	`escalations` table, queue, lead view	L1 reminder at 50 % of the step window; L2 supervisor at breach; L3 clinical lead at 1.5×; L4 director at 2× (T0/T1 only); one consolidated alert per level when the pathway target is missed (`kareshafaa/alert_policy.py`)
FHIR R4 resources	`GET /api/cases/{id}/fhir` (Bundle `collection`)	Patient, Organization, Practitioner(Role), DiagnosticReport, Observation, Task (case and steps), Communication, Consent, AuditEvent; capability at `/fhir/metadata`. Validated in tests with `fhir.resources`
Audit	`GET /api/audit`, `GET /api/audit/verify`	Every request and action; hash chain verified end to end
Reports	`GET /api/analytics/impact` (JSON/CSV)	Loop closure, time to specialist by tier, lost to follow-up, recall uptake, equity, "why loops fail" barriers – small cells suppressed

Worked example (produced by `scripts/handover_example.sh`)

- Inputs: `examples/input_oru_r01_fit_positive.hl7`, `examples/input_fhir_bundle_hba1c.json` (the script points the sample MRNs at two freshly seeded synthetic patients).
- Run record: `examples/example_run_record.md` (human-readable) and `examples/example_run_record.json`.
- FHIR output of one case: `examples/example_output_fhir_bundle_case.json`.

In that run: FIT 250 µg/g → rule `CRC-FIT-POS`, tier high, colonoscopy pathway with a 31-day target, ACK `MSA|AA`; HbA1c 8.1 % → rule `DM-A1C-DIAB`, tier high, confirm-then-GP pathway. The sweep at +1 day raised L1 and L2 (step breached); at +40 days a consolidated pathway-target alert. 25 FHIR resources validated;

audit chain valid; 20,000 rule evaluations agreed with the independent reference (0 safety-critical disagreements).

4. Test results

All results are on **synthetic data**. Every figure below is copied from a generated report in `docs/` or from a run made on 4 Nov 2026 (IST); the source is named next to each table. Nothing here is real-world evidence.

Automated test suite

Item	Result	Source
Test files	56 (`tests/test_*.py`)	repository
Tests run / passed	**3,606 / 3,606** (0 failed)	`pytest -q --cov=kareshafaa`, 4 Nov 2026
Line coverage	**88 %** of `kareshafaa/` (12,783 statements); `statemachine.py` 100 %, `clinical_rules.py` 99 %, `alert_policy.py` 98 %, `services/ingest.py` 95 %, `fhir.py` 93 %, `services/escalation.py` 92 %	same run
FHIR conformance	Exported case Bundles and every entry validated against FHIR R4B models (`fhir.resources` 8.3) plus required status value sets; CapabilityStatement valid; sample input Bundle valid	`tests/test_fhir_conformance.py`
HL7 v2	ORU^R01 parse (MSH/PID/OBR/OBX), ACK echoes the control id, ingest refused without permission, adversarial and malformed messages	`tests/test_fhir_conformance.py`, `test_ingest*.py`
Translation completeness	22 patient-facing text tables: same keys, no empty strings, same placeholders in en/ar/hi/ur	`tests/test_translation_completeness.py`
Accessibility	axe-core 4.14 WCAG 2.1 A/AA on 28 page views (staff roles, patient app in en/ar/hi/ur) + keyboard focus check: **0 problems** (26 before round 2)	`scripts/a11y_check.py`, `scripts/ci_a11y.sh`
Security scans	`pip-audit`: no known vulnerabilities; `bandit`: 0 high, 0 medium after fixes (48 low reviewed)	`docs/security.md`
Example run from a fresh clone	`scripts/handover_example.sh` passes from `git clone` in a temp dir	see `examples/example_run_record.md`

Baseline comparisons

Rules engine vs an independent reference implementation

`docs/differential_results.md`: 10,000,000 randomised results (boundary-weighted, all four programmes, malformed input, units) – **0 safety-critical disagreements** (one-sided 95 % upper bound 3.0×10^{-7}), 0 non-critical. Re-run by the handover script on 20,000 results with seed 20262020: 0 disagreements.

Alerting: earlier policy vs current policy

`docs/alert_precision.md` (5,930-case cohort, truth labelled independently of the alert code):

Metric	Earlier policy (Iteration 6)	Current policy
True breaches detected within one sweep	2,803 of 3,120 (recall 89.84 %)	3,120 of 3,120 (100 %)
Alerts L2+ / false positives	6,115 / 1,355 (precision 77.84 %)	5,104 / 0 (100 %)
Escalations reaching the expected level	476 of 841	841 of 841

No system (manual follow-up) vs KareShafaa – simulation

`docs/simulation_20k_results.json`: discrete-event simulation, 20,000 synthetic patients, 5 replications, seed 20261120, 120-day horizon. Values are the mean with the range across replications.

> **Read this carefully.** The "with KareShafaa" arm uses *assumed* parameters (for example notification median 1 day vs 6, no-show 11 % vs 18 %, closure documented 97 % vs 55 %; full list in the JSON `assumptions`). The difference between the arms is therefore an **assumption carried through a model, not a measured effect**. It shows what the assumptions imply end to end; the pilot must measure the real effect.

KPI	No system (baseline)	With KareShafaa
Median days to patient notified	8.0 (7.9–8.1)	1.3 (1.3–1.3)
Median days to patient understood	12.1 (12.0–12.1)	4.9 (4.9–4.9)
Median days to care reached	48.3 (47.7–48.7)	24.6 (24.4–24.7)
% notified	82.3 (81.9–82.6)	98.5 (98.4–98.5)
% understood (teach-back)	44.1 (43.9–44.3)	95.0 (94.8–95.2)
% appointment attended	73.2 (72.8–73.7)	95.9 (95.8–96.3)
% care reached or resolved	34.8 (34.6–35.2)	82.3 (81.7–82.8)
% loop closed with documentation	20.4 (20.3–20.6)	85.1 (84.6–85.4)
% still open/unresolved at 120 days	79.6 (79.4–79.7)	14.9 (14.6–15.4)
% FIT-positive reaching colonoscopy	29.3 (27.9–30.2)	88.0 (87.2–89.0)
Median days FIT-positive to colonoscopy	72.9 (70.5–77.1)	28.5 (28.2–28.8)
% review-flagged results actioned	80.2 (79.4–80.8)	100.0 (100.0–100.0)

ML outreach-priority model

`docs/ml_calibration.md` (temporal hold-out, 461 cases): logistic regression AUC **0.684**, ECE 0.036, calibration slope 1.007, 10/10 reliability bins within the 95 % CI, no fairness gate failed. The model only orders cases *within* an urgency tier; it never sets a tier or a decision. AUC 0.68 is modest – it is a work-ordering aid.

Robustness

Check	Result	Source
Edge-case cohort	20,000 planned people, stratified by programme × severity × scenario (missing/unknown values, unit mismatch, amended, duplicate identity, death mid-pathway, consent withdrawn, Ramadan, malformed HL7 …); every realised cell recorded, no result lost; 0 coverage gaps listed	`docs/edge_case_coverage.md`
Fault injection	54 deliberate faults in thresholds, boundaries and workflow – **54 caught** (100 %)	`docs/fault_injection_results.md`
Mutation testing	`clinical_rules.py` + `statemachine.py`: 1,560 mutants, **91.0 % killed**; the 12 real test gaps found were closed; the rest classified by hand	`docs/mutation_testing.md`
Load	Locust, 100 users, 4 workers: 3,502 requests, p95 69 ms, 1 failure	`docs/loadtest_locust.md`

Errors found and fixed (round 2)

- N+1 queries: recalls list 206 → 7 SQL statements; patient search 66 → 8; case detail 51 → 33 (Postgres 16).
- Accessibility: account menu accessible name did not match its visible text (every signed-in page); date inputs without a visible focus ring; one scrollable table not reachable by keyboard.
- Security: no CSP / HSTS / Permissions-Policy; bandit high (SHA-1 flag) and medium (f-string SQL in bootstrap).
- Clean-clone setup: the handover example needed `httpx` installed explicitly.

Known limits (honest)

- All data synthetic; no real-world outcome, accuracy or time-saving has been measured.
- Simulation gains come from assumed parameters (see above).
- Clinical thresholds and SLA hours are illustrative defaults marked **CONFIRM** until clinical sign-off (`docs/clinical_rules.md`, `kareshafaa/sla.py`).
- FHIR validated against base R4B models, not a UAE/Malaffi implementation guide; required value sets checked only for status fields.
- `script-src 'unsafe-inline'` remains in the CSP (inline scripts in templates).
- Patient messages are simulated (no SMS/WhatsApp gateway); no MFA or SSO.
- Free hosting is outside the UAE and sleeps when idle (cold start); only synthetic data is ever hosted there.

5. Pilot plan (condensed)

Full plan with sources: `docs/pilot_plan.md`. Status: **proposal** – no partner has agreed yet and no organisation named there has endorsed KareShafaa. Values marked ASSUMPTION or CONFIRM must be agreed.

Design

90 days in one Abu Dhabi primary care cluster, alongside current practice (staff keep existing referral routes; KareShafaa adds ownership, deadlines, patient messages and evidence of closure). Bowel (FIT) first; cervical and breast from day 45 if the go/no-go gate is passed. UAE-hosted only; free public hosting is never used for real data.

Participants

Group	Who (roles, not named people)
Review lead	**Programme Clinical Lead (to be named)** – owns the safety huddle, the gate and sign-off
Clinical users	Screening coordinators, family doctors and the receiving endoscopy / breast imaging units at the pilot sites
Patients	Adults with an abnormal screening result at the pilot sites (eligibility per programme specification; to confirm with the cluster). Volume **ASSUMPTION:** 300–600 abnormal results in 90 days
Governance	Regulator and programme owner contacts (DoH, ADPHC), the cluster's data protection and information security officers (to be named)
Technical	KareShafaa product lead and engineer (to be named); cluster IT for the laboratory HL7 feed
Patient voice	Patient and community representatives reviewing messages in Arabic, English, Hindi and Urdu

Success criteria at day 90 (from `docs/pilot_plan.md`)

1. No serious safety incident attributable to KareShafaa.
2. Loop closure and within-target rates better than baseline for bowel screening, with the 90 % CI reported.
3. No equity group more than 10 points (CONFIRM) below the overall rate without an explanation and action.
4. Coordinators and clinicians would keep using it (short survey, ≥ 70 % agree, ASSUMPTION).
5. Security assessment passed and audit chain intact for the whole pilot.

KPI targets (loop closure +20 points, ≥ 80 % within target for critical/urgent, lost to follow-up halved) are **ASSUMPTIONS** until agreed.

Review dates (relative to pilot start, day 1 = first live result)

When	Review	Led by
Day −56 to −1	Readiness: data sharing agreement, approvals, ADHICS assessment, clinical sign-off of rules/SLAs/messages, HL7 test feed	Programme Clinical Lead (to be named)
Days 1–14	Baseline extract (last 12 months); staff training	Clinical lead + cluster data team
Weekly from day 15	Safety huddle: every critical/urgent case and every human override	Programme Clinical Lead (to be named)
Day 30	First KPI and equity review	Clinical lead + product lead
Day 45	**Go/no-go gate** for cervical and breast	Programme Clinical Lead (to be named) + steering group
Day 60	Equity review by language, nationality group and region	Clinical lead
Day 90	Final report to the steering group; continue / extend / stop	Steering group

6. Handover session (60 minutes)

Attendees: Programme Clinical Lead (to be named), receiving technical owner (to be named), product lead, one coordinator representative. **Pre-read:** `README.md`, `test_results.md`, `examples/example_run_record.md`.

Agenda

Time	Item	Material
0–5	Purpose, scope in / out	`problem_brief.md`
5–20	**Results review**: test suite, rules vs reference, alert precision, simulation (and its assumptions), a11y and security, known limits	`test_results.md`
20–35	**Walkthrough**: run `scripts/handover_example.sh` live; then the app – coordinator queue, a case with "why this priority" and SLA timer, lead alerts, Loop impact incl. "Why loops fail", patient app in Arabic	`runnable_project.md`, `how_to_run_the_task.md`
35–48	**Q&A** (checklist below)	–
48–55	**Decisions to agree**	below
55–60	**What happens next** and sign-off	below

Q&A checklist

- [ ] Can the receiving team run the example from a clean clone without help?
- [ ] Where do secrets live, and who rotates `KARESHAFAA_SECRET_KEY`?
- [ ] Which values are CONFIRM (thresholds, SLA hours, equity flag level) and who signs them?
- [ ] How is a result with an unknown code or identity doubt handled? (human review queue)
- [ ] How is the audit chain verified, and how often?
- [ ] What happens when the app is overloaded or the database is slow? (503 + Retry-After; nothing half-saved)
- [ ] What does the ML model do and not do? (orders within a tier only)
- [ ] Which data protection mappings are the team's reading rather than legal advice?
- [ ] What is simulated (patient messaging, simulation gains)?

Decisions to agree

#	Decision	Options	Agreed
1	Pilot site cluster and start date	–	
2	Sign-off owner for rules, SLA hours and patient messages	Programme Clinical Lead (to be named)	
3	KPI targets (replace ASSUMPTIONS)	as proposed / revised	

#	Decision	Options	Agreed
4	Hosting for the pilot	UAE cloud region / cluster data centre	
5	Messaging gateway (SMS / WhatsApp)	–	
6	Follow-up work priority: CSP nonces, MFA, external pen test, Malaffi	order	

What happens next

1. Receiving team reruns the example and the full test suite on their own machine (target: within 1 week).
2. Clinical sign-off workshop for CONFIRM values.
3. Readiness steps in `pilot_plan.md` (day −56 to −1).
4. Agree the date of the first review (day 30 after go-live).

Sign-off

Role	Name	Accepted (yes / with conditions)	Date	Signature
Programme Clinical Lead (to be named)				
Receiving technical owner (to be named)				
KareShafaa product lead				

KareShafaa handover example: run record

Synthetic data only. Seed 2026, 300 synthetic patients, commit `2a6e098`, started 2026-11-04T02:48:18Z, 12.3 s.

Input: HL7 v2 ORU^R01 (`docs/handover/examples/input_oru_r01_fit_positive.hl7`)

- Ingest status: **processed**; actionable: **True**; message id `LAB-20261020-000123`
- Alert: case 306, Bowel (colorectal) screening - FIT, tier **high**, rule `CRC-FIT-POS`
- Pathway: colonoscopy; target Colonoscopy within 31 days of referral (DoH) (due 2026-12-05T02:48:18.998507)
- State now: Result issued; human review required: True
- SLA timer on the current step: {"status": "green", "fraction": 0.0, "due_at": "2026-11-05T02:48:18.998507", "escalation_level": 0}
- Sweep at +1 days: {"checked": 1, "reminders": 1, "supervisor": 1, "breaches": 1}; escalation levels so far [1, 2]
- Sweep at +40 days: {"checked": 1, "supervisor": 1, "target_breach_alerts": 1}; escalation levels so far [1, 2, 2]
- FHIR export: `case_306_fhir_bundle.json`

HL7 ACK returned:

```
MSH|^~\&|KARESHAFAA|PROGRAMME|LAB|LAB|20261104024819+0000||ACK^R01|ACKLAB-20261020-000123|P|2.
5.1
MSA|AA|LAB-20261020-000123|accepted
```

Input: FHIR R4 Bundle (`docs/handover/examples/input_fhir_bundle_hba1c.json`)

- Ingest status: **processed**; actionable: **True**; message id `FHIR-LAB-20261020-000777`
- Alert: case 307, Diabetes screening - HbA1c / fasting glucose, tier **high**, rule `DM-A1C-DIAB`
- Pathway: confirm_then_gp; target Confirmatory test within 4 weeks (CONFIRM) (due 2026-12-02T02:48:19.033506)
- State now: Result issued; human review required: True
- SLA timer on the current step: {"status": "green", "fraction": 0.0, "due_at": "2026-11-05T02:48:19.033506", "escalation_level": 0}
- Sweep at +1 days: {"checked": 1, "reminders": 1, "supervisor": 1, "breaches": 1}; escalation levels so far [1, 2]
- Sweep at +40 days: {"checked": 1, "supervisor": 1, "target_breach_alerts": 1}; escalation levels so far [1, 2, 2]
- FHIR export: `case_307_fhir_bundle.json`

Checks

- FHIR: 50 resources validated (AuditEvent, Communication, Consent, DiagnosticReport, Observation, Organization, Patient, Practitioner, PractitionerRole, Task) - valid: **True**
- Audit chain: {"valid": true, "checked": 5617, "head": "35ed6fed439dfc6d8e23475c2b091296d3ce191ef2959a6ddfa98a191f96908e"}

- Rules engine vs independent reference: 20,000 randomised results (seed 20262020): agree 20,000, safety-critical disagreements **0** (95 % upper bound 1.5e-04)